Spoken Tutorial – Understanding AI, Machine Learning, and Deep Learning

Title: Understanding AI, Machine Learning, and Deep Learning

Module	Introduction to AI Concepts
Episode	AI, ML, and Deep Learning Explained
Learning Objective	At the end of this tutorial learner will be able to 1. Differentiate between Artificial Intelligence, Machine Learning, and Deep Learning.
Approx. Duration	10 minutes
Outline	• What is Artificial Intelligence (AI)? • Understanding Machine Learning (ML) • Exploring Deep Learning • Key Differences
Meta Tags	AI, Artificial Intelligence, Machine Learning, ML, Deep Learning, DL, Neural Networks, Introduction to AI
Pre-requisite Tutorial	Basic computer literacy.

Script

Visual Cue	Narration
	Welcome to this **Spoken Tutorial**. We will learn about **Artificial Intelligence**. We will also learn about **Machine Learning**. Finally, we will cover **Deep Learning**.
	After this **tutorial**, you will know the difference. You will understand **Artificial Intelligence**. You will also understand **Machine Learning**. And you will understand **Deep Learning**.
	To follow this **tutorial**, you need a computer. You also need an internet connection.
	You should know basic computer use. You do not need to know **AI** or programming.
A futuristic robot brain with gears and light, representing intelligence and complex thinking.	Have you seen how your phone suggests words? How do apps recommend movies? This is **Artificial Intelligence**, or **AI**. **AI** is about making computers smart. They do tasks like humans.

A human brain connected with circuits and digital patterns, showing various intelligent tasks like language, vision, and problem-solving icons around it.	**AI** makes computers intelligent. For example, computers can understand language. They can recognize faces. They can solve hard math problems. This means machines can think. They can act like humans.
A robot hand pointing to a chart with data points, showing a learning curve, representing pattern recognition.	Next, let's learn about **Machine Learning**, or **ML**. **ML** is a part of **AI**. Here, computers learn from data. They find patterns. We do not tell them every rule.
A programmer typing IF-THEN rules on a screen, with a simple illustration of a cat next to it, showing explicit instructions.	Let's see an example. In normal programming, you give exact rules. For a cat, you might say: 'IF pointy ears AND whiskers, THEN it is a cat'. You write every rule yourself.
A computer screen displaying many diverse images of cats and non-cats, with a progress bar indicating the machine learning process.	But in **Machine Learning**, it is different. You show the computer many pictures of 'cats' and 'not cats'. The computer then learns what a cat looks like by itself. It finds the rules from the data.
A complex neural network diagram with multiple layers of interconnected nodes, resembling brain cells, glowing with data flow.	Now, let's look at **Deep Learning**. It is a special type of **Machine Learning**. **Deep Learning** uses **neural networks** like our brain. These **networks** have many layers. So they are called 'deep'.
A futuristic car driving autonomously on a city street, with overlay graphics showing object recognition (pedestrians, traffic lights), and a voice assistant icon.	**Deep Learning** helps computers learn very complex things. It learns from a lot of data. For example, self-driving cars use it to see people. Voice assistants use it to understand what you say. Even **AI** that makes images uses **Deep Learning**.
A Venn diagram showing three concentric circles: the largest labeled 'AI', the middle labeled 'Machine Learning', and the innermost labeled 'Deep Learning'.	Let's quickly review. **AI** is the main idea of making machines smart. **Machine Learning** is a part of **AI**. Machines learn from data in **Machine Learning**. **Deep Learning** is a special part of **ML**. It uses deep **networks** to find complex patterns.
	In this **tutorial**, you learned the difference. We discussed **Artificial Intelligence**. We also discussed **Machine Learning**. And we discussed **Deep Learning**. We saw their meanings. We also saw examples.
	Your assignment is to find one more real-world example. This applies to **AI**, **Machine Learning**, and **Deep Learning**. Talk about these examples with a friend.

	This **Spoken Tutorial** is from **EduPyramids Educational Services Private Limited**. It is from **SINE**, **IIT Bombay.** Thank you for watching!